

26 Matrix Maximum size rectangle

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#include <bits/stdc++.h>
using namespace std;
class Solution {
public:
    // Largest rectangle area in histogram
    int largestRectangleHistogram(vector<int>& heights) {
        int n = heights.size();
        stack<int> st;
        int maxArea = 0;
        for (int i = 0; i <= n; i++) {
            int cur = (i == n ? 0 : heights[i]);    while (!st.empty() && cur < heights[st.top()]) {
                int h = heights[st.top()];
                st.pop();
                int left = st.empty() ? 0 : st.top() + 1;
                int width = i - left;
                maxArea = max(maxArea, h * width);
            }
            st.push(i);
        }
        return maxArea;
    } // Main required function by GFG → MUST be named maxArea
    int maxArea(vector<vector<int>>& mat) {
        int n = mat.size();
        int m = mat[0].size();
        vector<int> heights(m, 0);
        int maxRect = 0;
        for (int i = 0; i < n; i++) { // Build/update histogram height
            for (int j = 0; j < m; j++) {
                if (mat[i][j] == 1) heights[j]++;
                else heights[j] = 0;
            } // Evaluate largest rectangle for this histogram
            maxRect = max(maxRect, largestRectangleHistogram(heights));
        }
        return maxRect;
    }
};
```

The screenshot displays a coding platform interface with a dark theme. On the left, the 'Output Window' shows 'Compilation Results' and a green checkmark indicating 'Problem Solved Successfully'. Below this, statistics are listed: 'Test Cases Passed: 1115 / 1115', 'Attempts: Correct / Total: 1 / 1', 'Accuracy: 100%', 'Points Scored: 8 / 8', and 'Your Total Score: 94'. At the bottom, 'Solve Next' buttons are visible for 'Matrix Chain Multiplication', 'Longest Increasing Path in a Matrix', and 'Largest rectangular sub-matrix whose sum is 0'. The main editor on the right shows C++ code for the 'Matrix Maximum size rectangle' problem. The code includes a class 'Solution' with two methods: 'largestRectangleHistogram' and 'maxArea'. The 'maxArea' method iterates through each row of the matrix, builds a histogram of heights, and calls 'largestRectangleHistogram' to find the maximum area for that row. The overall maximum area is then returned.